Networking Basics for DevOps

10 beginner-friendly questions and answers

1. What is Networking?

Networking is the process of connecting computers, servers, and other devices so they can communicate and share data or resources. For example, computers connected to the same office Wi-Fi form a network.

2. Explain LAN, WAN, and Internet.

LAN (Local Area Network): Connects devices within a small area, such as a home, office, or school.
WAN (Wide Area Network): Connects networks across large geographical areas. A company may use a WAN to connect offices in different cities.
Internet: A worldwide network of interconnected networks that lets devices and services communicate.

3. What is the OSI Model?

The OSI (Open Systems Interconnection) Model is a seven-layer reference model used to understand how data moves between devices over a network. It helps people learn networking and troubleshoot communication problems.

4. Explain the 7 layers of the OSI Model.

7. Application: Provides network services to applications, such as web and email services.
6. Presentation: Handles data formatting, encryption, and compression.
5. Session: Establishes, manages, and ends communication sessions.
4. Transport: Delivers data between applications; TCP and UDP operate here.
3. Network: Handles logical addressing and routing; IP operates here.
2. Data Link: Transfers frames over a local link and uses MAC addresses.
1. Physical: Sends raw bits through cables, radio signals, or other physical media.

Memory tip (Layer 7 to 1): All People Seem To Need Data Processing.

5. What is the TCP/IP Model?

The TCP/IP Model describes the protocols used for communication across networks, including the Internet. It is commonly shown with four layers:
Application: HTTP/HTTPS, DNS, SSH, and other application protocols.
Transport: TCP and UDP.
Internet: IP and ICMP; handles addressing and routing between networks.
Network Access (Link): Local network delivery using technologies such as Ethernet and Wi-Fi.

6. Difference between OSI and TCP/IP.

OSI: A seven-layer reference model that is mainly used to explain and troubleshoot networking concepts.
TCP/IP: A practical protocol suite and model used for real-world Internet communication, commonly represented with four layers.
The OSI model separates Presentation and Session layers, while TCP/IP generally includes their functions in the Application layer. The OSI Data Link and Physical layers are commonly combined as the TCP/IP Network Access layer.

7. What is an IP Address?

An IP (Internet Protocol) address is a logical address used to identify an interface on an IP network and help deliver packets to the correct destination. IPv4 addresses look like `192.168.1.10`; IPv6 addresses are longer and use hexadecimal notation. IP addresses can be private or public.

8. What is DNS?

DNS (Domain Name System) translates domain names, such as `example.com`, into IP addresses that computers use to communicate. You can use `nslookup example.com` to query DNS and check the result.

9. What is ICMP?

ICMP (Internet Control Message Protocol) is used by network devices to send error reports and diagnostic information. The `ping` command commonly uses ICMP Echo Request and Echo Reply messages to test reachability. A failed ping does not always mean a device is down, because firewalls may block ICMP.

10. Why is Networking important in DevOps?

Networking helps DevOps teams connect applications, servers, containers, cloud services, and CI/CD tools. It is important for:

- Deploying and accessing applications.
- Configuring DNS, IP addresses, ports, firewalls, and load balancers.
- Connecting containers, Kubernetes clusters, and cloud resources.
- Troubleshooting connectivity, latency, and service outages.
- Securing communication between systems.

Quick revision: LAN = small local network; WAN = network across large areas; DNS = name-to-IP lookup; IP = addressing and routing; ICMP = network diagnostics and error messages.